

"PAPER_{0:D3}" -f [int]# PAPER #119 ■ UQFF 7-System Equation Reference: Complete Variable Tables

Author: Daniel T. Murphy

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Title: UQFF General Equation Systems ■ Compressed, Resonant, Buoyancy, Superconductive, Triadic, Quadratic, and Master Buoyancy with Full Variable Equations

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ($\tau = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$, $\mathbf{i} = 0.61$) **Date:** March 2026 **Source:** Grok thread 2fe4fa3e (DeepSearch extraction, Sept 22, 2025) **Index Slot:** ■1.16 **UQFF Equation Systems Reference** **Preceding Paper:** PAPER064 (4 simplified modes, Batch 23)

Abstract

The Unified Quantum Field Superconductive Framework (UQFF) operates in seven distinct equation systems, each derived from the master unified field F_U but applied at different scales and computational modes. This paper provides a complete variable-equation reference for all seven systems: (1) UQFF Compressed ■ the compact unified form; (2) UQFF Resonant ■ oscillatory/resonant terms; (3) UQFF Buoyancy ■ the Ubi opposition system; (4) UQFF Superconductive ■ [SCm] reactivity coupling; (5) UQFF Triadic ■ triadic master equations for plasma/turbulence systems; (6) UQFF Quadratic ■ quadratic field approximations; and (7) UQFF Master Buoyancy ■ extended Ubi with Mayan-aligned time scaling. All variable equations are listed with their definitions, dimensions, and physical origins. This reference supersedes and extends PAPER064 (four simplified modes) with the full-complexity forms extracted from the 393-page verification corpus.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. Master Unified Field Equation F_U

Before defining the seven systems, the master equation from which all are derived:

$$F_U = \sum_{i=1}^4 \left[k_i U_{gi} - \beta_i U_{gi} \cdot \frac{\omega_g M_{bh}}{d_g} \cdot E_{react} \right] + \sum_j \left[\frac{\mu_j}{r_j} \left(1 - e^{-\gamma t} \cos(\pi t_n) \right) \hat{\phi}_j \right] + \left(g_{\mu\nu} + \eta T_s^{\mu\nu} \right) - \sum_i \left[\delta_i U_i E_{react} \right] + \text{CRP}: \sum D_E \frac{\partial^2 n}{\partial p^2} e^{-\gamma t}$$

2. System 1: UQFF Compressed (Compact Unified Form)

Long Form:

$$F_{\{U, \text{Compressed}\}} = \sum_{i=1}^4 \left[k_i U_{\{gi\}} - \beta_i U_{\{gi\}} \frac{\omega_g M_{\{bh\}}}{d_g} E_{\{react\}} \right] + \sum_j \frac{\mu_j}{r_j} (1 - e^{-\gamma t \cos(\pi t_n)}) \hat{\phi}_j + (g_{\mu\nu} + \eta T_s^{\mu\nu}) - \sum_i \delta_i U_i E_{\{react\}} + \sum D_E \frac{\partial^2 n}{\partial p^2} e^{-\gamma t}$$

Variable Equations:

Symbol	Value / Equation	Units	Description
$F_{\{U\}}$	Full sum of components	J/m ⁴	Unified energy density
i	1-4		Gravity range index
k_i	$k_1=1.5, \backslash; k_2=1.2, \backslash; k_3=1.8, \backslash; k_4=1.0$		Coupling constants
$U_{\{g1\}}$	$k_1 \mu_s (M_s/r) e^{-\alpha t} \cos(\pi t_n) (1+\beta \eta_{\{def\}})$	N/m ⁴	Dipole gravity
$U_{\{g2\}}$	$k_2 (\lambda_{\{vac, [UA]\}} + \lambda_{\{vac, [SCm]\}}) (M_s/r^2) \backslash, S(r-R_b) (1+\delta_{\{sw\}} v_{\{sw\}}) H_{\{SCm\}} E_{\{react\}}$	N/m ⁴	Bubble gravity
$U_{\{g3\}}$	$k_3 \sum_j B_j \backslash \cos(\omega_s t) \backslash, P_{\{core\}} \backslash, E_{\{react\}}$	N/m ⁴	Disk strings gravity
$U_{\{g4\}}$	$k_4 \lambda_{\{vac, [SCm]\}} (M_{\{bh\}}/d_g) e^{-\alpha t} \cos(\pi t_n) (1+f_{\{feedback\}})$	N/m ⁴	BH-star interaction
$\backslash \beta_i$	0.61 (uniform i=1-4)		Buoyancy coupling
$\backslash \omega_g$	7.3×10^{-16} rad/s	rad/s	Galactic spin (V=220 km/s, r=8 kpc)
$M_{\{bh\}}$	8.15×10^{36} kg	kg	BH mass (4.1-106 M?, Sgr A*)
d_g	2.55×10^{20} m	m	Galactic distance (27,000 ly)
$E_{\{react\}}$	$10^{46} e^{-0.0005 t} = \rho_{\{vac, [SCm]\}} v_{\{SCm\}}^2 / \rho_{\{vac, A\}} \cdot e^{-\kappa t}$	W/m ⁴	[SCm] reactor efficiency
$\backslash \mu_j$	$(10^3 + 0.4 \sin(\omega_c t)) \times 3.38 \times 10^{20}$	T-m ⁴	Magnetic moment of string j
r_j	1.496×10^{13} m	m	String distance (100 AU)
$\backslash \gamma$	0.00005 day? ⁴	day? ⁴	String decay rate ($\tau \approx 55$ yr)
t_n	$t - t_0 < 0$ for TRZs	s or days	Negative time
$\hat{\phi}_j$	≈ 1		String unit vector

Symbol	Value / Equation	Units	Description
$g_{\mu\nu}$	$\text{diag}(1, -1, -1, -1)$	■	Minkowski metric
η	1×10^{-22}	■	Aether coupling
$T_s^{\mu\nu}$	$\approx 1.123 \times 10^7 \text{ J/m}^3$	J/m^3	Stress-energy tensor
δ_i	1.0	■	Inertial coupling
U_i	$\lambda_i \rho_{\text{vac}} [\text{SCm}] \rho_{\text{vac}} [\text{UA}] \omega_s(t) \cos(\pi t_n) (1 + f_{\text{TRZ}})$	J/m^3	Universal inertia
D_E	$\propto E^{0.5}$ (Kolmogorov)	■	CRP diffusion coefficient
$n(p)$	$p^{-2.2} e^{-p/p_{\text{max}}}$	$m^3 (\text{GeV}/c)^3$	CRP momentum distribution

3. System 2: UQFF Resonant (Oscillatory Terms)

Physical basis: Models oscillating vacuum fields via $\cos(\pi t_n)$ driving TRZs and negentropy.

Core resonant components:

```
\text{Resonant modulator: } \cos(\pi t_n) = \cos(\pi(t - t_0))
\text{String buildup: } (1 - e^{-\gamma t} \cos(\pi t_n))
\text{Reactivity decay: } e^{-\kappa t} \text{ with } \kappa = 0.0005 \text{ day}^{-1}
```

Variable Equations:

Symbol	Value / Equation	Units	Description
$\cos(\pi t_n)$	$\cos(\pi(t-t_0))$	■	Periodic reversal oscillator, period = 2 days
t_0	Initial epoch	days	Reference time
t_n	$t - t_0 < 0$	days	Negative time in TRZs
γ	0.00005 day^{-1}	day^{-1}	Decay rate, $\tau = 1/\gamma \approx 54.8 \text{ yr}$
κ	0.0005 day^{-1}	day^{-1}	Reactivity decay, $\tau_{\kappa} \approx 5.48 \text{ yr}$
$(1 - e^{-\gamma t} \cos(\pi t_n))$	$\approx \gamma t \cos(\pi t_n)$ for small t	■	Resonant buildup in U_m
f_{TRZ}	0.1	■	TRZ factor scaling U_i for negentropy
ω	$\pi \text{ rad/s (rad/day)}$	rad/day	Cycle constant

Application: EP-09 (3C 273 jet ratio $R = |\cos(\pi t_{n1})|/|\cos(\pi t_{n2})| > 100$ for $\Delta t \approx 0.5 \text{ days}$).

4. System 3: UQFF Buoyancy (U_{b,i} Opposition System)

Long Form:

$$U_{b,i} = -\beta_i U_{g,i} \cdot \frac{\omega_g M_{bh}}{\Delta_{sw} \lambda_{vac,sw}} \cdot [UA] \cdot \cos(\pi t_n)$$

Variable Equations:

Symbol	Value / Equation	Units	Description
$U_{b,i}$	Full expression above	J/m ³	Buoyancy density (opposes $U_{g,i}$)
β_i	0.61		Buoyancy coupling (uniform $i=1-4$)
$U_{g,i}$	See System 1	J/m ³	Gravity component
ω_g	7.3×10^{-16} rad/s	rad/s	Galactic rotation rate
M_{bh}	8.15×10^{36} kg	kg	SMBH mass (Sgr A*)
d_g	2.55×10^{20} m	m	Galactic distance
Δ_{sw}	0.01		Solar wind modulation factor
$\lambda_{vac,sw}$	$\frac{\rho_{sw}}{c^2} \approx 7.2 \times 10^{-4}$ J/m ³	J/m ³	Wind vacuum density
[UA]	10^{-11} C	C	Trapped Aether charge
$\cos(\pi t_n)$	Oscillator		Time-reversal modulation

Physical interpretation: $U_{b,i}$ feeds outflows by opposing gravity. For BNS mergers (EP-11, GW170817): dynamical ejecta fraction $\approx 1 - \beta_i \approx 0.39 \approx 40\%$. For jets: $\cos(\pi t_n)$ reversals create $R > 100:1$ asymmetry (EP-09).

5. System 4: UQFF Superconductive ([SC_m] Reactivity)

Long Form (dual form):

$$E_{react} = 10^{46} e^{-0.0005 t} \equiv \frac{\rho_{vac,[SC_m]} v_{SC_m}^2}{\rho_{vac,A}} e^{-\kappa t}$$

Variable Equations:

Symbol	Value / Equation	Units	Description
E_{react}	$10^{46} e^{-\kappa t}$	W/m ³	[SC _m] superconducting reactivity
$\rho_{vac,[SC_m]}$	7.09×10^{-37} J/m ³	J/m ³	SC _m vacuum energy density
v_{SC_m}	1×10^8 m/s (= c/3)	m/s	SC _m propagation velocity
$\rho_{vac,A}$	1×10^{-23} J/m ³	J/m ³	Aether vacuum density
κ	0.0005 day ⁻¹	day ⁻¹	Reactivity decay constant

Symbol	Value / Equation	Units	Description
10^{46}	$= \frac{\rho_{\text{vac}} [\text{SCm}]}{v_{\text{SCm}}^2 / \rho_{\text{vac}} A}$	W/m ²	Quasar-scale base efficiency
[SCm]	10^{15}	m ²	Superconductive material density
[UA]	10^{-11}	C	Universal Aether charge

Calibration: EP-05 (Fermi LAT 4LAC blazars): $\bar{\kappa} = 0.000497 \pm 5\% \text{ day}^{-1}$ across 3,743 sources. EP-07 (Parker Solar Probe): $\delta_{\text{sw}} = 0.01 = [\text{SSq}]/57$.

6. System 5: UQFF Triadic (Triadic Master Equations)

Long Form (e.g., Westerlund 2 plasma turbulence):

$$F_{\{U, \text{Triadic}\}} = F_U + \left(U_{g3} \cdot U_{b,i} \cdot U_m \right)^{1/3} \cdot \exp \left(- [\text{SSq}] \cdot \frac{n}{26} \right)$$

Variable Equations:

Symbol	Value / Equation	Units	Description
$F_{\{U, \text{Triadic}\}}$	$F_U +$ geometric mean term	J/m ²	Triadic unified field
$(U_{g3} \cdot U_{b,i} \cdot U_m)^{1/3}$	Geometric mean of disk gravity, buoyancy, magnetism	J/m ²	Triadic coupling
[SSq]	$\log_{10}(\frac{\rho_{\text{vac}}}{\lambda_{\text{vac}}}) \approx 38$ (for 10^{-38} ratios)		Self-similar quotient
n	$1 \leq 26$ (level index)		UQFF ladder level
n = 13	Plasma systems (Westerlund 2, Pillars)		Triadic plasma level
$\exp(-[\text{SSq}] \cdot n/26)$	$\exp(-38 \cdot n/26)$		Level-scaled suppression
$Q_{\text{wave}, 47}$	Mean: 3.97×10^4 J/m ² , std: 5.11×10^4 J/m ²	J/m ²	47-system wave energy density
Jarque-Bera	8.78 (p=0.012)		Non-normality statistic for Q_{wave}
Leptokurtosis	0.037		Excess kurtosis of Q_{wave} distribution

Systems using Triadic: Westerlund 2 (turbulence), Pillars of Creation (buoyancy), TOI 1227 b (mass loss $\approx 10^{12}$ g/s), PLCK G287.0+32.9 (double relics), ASKAP J1832-0911, PSZ2 G181.06+48.47, AT2024tvd, G359.13142-0.20005.

7. System 6: UQFF Quadratic (Quadratic Field Approximations)

Long Form (low-order approximation of potential):

$$V(r) \approx a_0 + a_1 r + a_2 r^2$$

$T_{s^{\mu\nu}} = 1.27 \times 10^3 + 1.11 \times 10^7 \text{ J/m}^3 \text{ (quadratic sum)}$

Variable Equations:

Symbol	Value / Equation	Units	Description
$V(r)$	$\sum_n a_n r^n$ $\approx$ quadratic for low deg	J or m/s	Field potential approximation
a_0	Constant term		Polynomial fit coefficient 0
a_1	Linear coefficient		Polynomial fit coefficient 1
a_2	Quadratic coefficient ($\approx 10^{-12}$ for n=8)		Polynomial fit coefficient 2
R^2	≈ 0.95 (for ENSDF n=8 bindings)		Quadratic fit quality
$T_{s^{\mu\nu}}$	$1.27 \times 10^3 + 1.11 \times 10^7$	J/m	Stress tensor quadratic sum
$a_2^{\{Pb-206\}}$	$\approx 10^{-12}$		Fit for EP-04 Pb-206 n=8 levels

Application: Nuclear binding energy polynomial fits (EP-04 ENSDF Pb-206, EP-02 PDG 2025) use degree-8 polynomials that reduce to effective quadratic forms for low-n regimes.

8. System 7: UQFF Master Buoyancy (Extended $U_{b,i}$ with Mayan Alignment)

Long Form:

$$U_{b, \text{Master}} = -\beta_i U_{g,i} \frac{\omega_g}{M_{bh}\{d_g\}(1+\delta_{sw})\lambda_{vac,sw}} [UA] \cos(\pi t_n) + \exp(-(p-t)) \cdot \frac{U_m}{\rho_{vac,UA}}$$

Physical basis: Extends standard $U_{b,i}$ with a time-scaled magnetic feed term. The $\exp(-(p-t))$ factor aligns with Mayan calendar time constants (Baktun cycle) in the UQFF cosmological scaling.

Variable Equations:

Symbol	Value / Equation	Units	Description
$U_{b, \text{Master}}$	Full expression above	J/m	Master buoyancy extended form
Standard $U_{b,i}$	System 3 above	J/m	Base buoyancy term
$\exp(-(\pi - t))$	For $t < \pi$ days, decays toward Euler		Mayan/Master time alignment factor
U_m	$\sum_j [\mu_j/r_j(1 - e^{-\gamma t \cos(\pi t_n)}) \hat{\phi}_j] P_{SCm} E_{react}$	J/m	Universal magnetism (lossless strings)
$\rho_{vac,UA}$	7.09×10^{-36} J/m	J/m	UA vacuum energy density
P_{SCm}	≈ 1 (Sun)		SCm core penetration factor

Symbol	Value / Equation	Units	Description
E_{react}	$10^{46} \text{ e}^{-\kappa t}$	W/m ²	Reactor efficiency

****Distinction from PAPER_064:**** PAPER_064 documents the four simplified operational modes. This Master Buoyancy form adds the resonant magnetic feed term $U_m / \rho_{\text{vac,UA}}$, which drives enhanced negentropy in long-period TRZ cycles spanning Mayan-scale time constants (Baktun = 394 yr $\approx 1/\kappa^{0.33}$).

9. Shared Constants Across All Seven Systems

Constant	Value	Physical Meaning
κ	0.0005 day ⁻¹	E_{react} decay rate (EP-05 Fermi LAT)
[SSq]	0.57	Self-similar quotient (EP-04 Pb-206 S _n)
β_i	0.61	Buoyancy coupling (EP-10 IceCube, EP-11 GW170817)
F_{rel}	$4.31 \times 10^{33} \text{ N}$	Relativistic unified force scale
k_{η}	10^{-113}	LENR exponential damping factor
ω	$\pi \text{ rad/day}$	UQFF cycle constant
α	0.001 day ⁻¹	Dipole/BH time decay in U_{g1} , U_{g4}
δ_{sw}	$0.01 = [\text{SSq}]/57$	Solar wind modulation (EP-07 PSP)
v_{sw}	$5 \times 10^5 \text{ m/s}$	Solar wind velocity (EP-07)
H_{SCm}	≈ 1	Heliosphere SCm factor
f_{feedback}	0.1	AGN/BH feedback factor
f_{TRZ}	0.1	TRZ negentropy factor
λ_i	1.0	Inertial coupling for U_i
p_{max}	10^{16} eV	CRP momentum cutoff (EP-10 IceCube)
γ_{CRP}	0.00005 day ⁻¹	CRP decay rate ($\tau \approx 55 \text{ yr}$)

10. Cross-Reference to Implemented Code

System	Source Code	Class/Function	PAPER_064 equiv.
Compressed	source2.cpp L1960, add_uqff_to_8_models.py L24	UQFF_Compressed	Mode 1 ?
Resonant	source2.cpp L1961, add_uqff_methods.py	UQFF_Resonant	Mode 2 ?
Buoyancy	add_uqff_to_8_models.py L68	F_U_Bi_i / UQFFMasterBuoyant	Mode 3 ?
Superconductive	add_uqff_to_8_models.py L48	UQFF_Superconductive	Mode 4 ?

System	Source Code	Class/Function	PAPER_064 equiv.
Triadic	add_uqff_to_8_models.py L76, add_uqff_methods.py L226	UQFF_Triadic	■ (new)
Quadratic	add_uqff_to_8_models.py L90, add_uqff_methods.py L291	UQFF_Quadratic	■ (new)
Master Buoyancy	add_uqff_to_8_models.py L68	UQFFMasterBuoyant	■ (new)

11. Summary Table

System	Core Equation	Key Parameter	Primary Verification
Compressed	$F_U = \Sigma k_i U_{gi} - \beta_i U_{gi} \omega_g M_{bh}/d_g E_{react}$	$\kappa = 0.0005 \text{ day?}$ ■	EP-05 Fermi LAT
Resonant	$\cos(\pi t_n), (1 - e^{-\gamma t} \cos(\pi t_n))$	Period = 2 days	EP-09 3C 273
Buoyancy	$U_{b,i} = -\beta_i U_{g,i} \omega_g M_{bh}/d_g (1 + \delta t_{sw} \lambda_{vac, sw}) [UA] \cos(\pi t_n)$	$\beta_i = 0.61$	EP-11 GW170817
Superconductive	$E_{react} = 10^{46} e^{-\kappa t}$	$\kappa = 0.0005 \text{ day?}$ ■, $v_{SC} = c/3$	EP-08 JCAP
Triadic	$F_{U, tri} = F_U + (U_{g3} U_{b,i} U_m)^{1/3} e^{-[SSq] n/26}$	$[SSq] = 0.57, n = 13$	Qwave47 mean
Quadratic	$V(r) \approx a_0 + a_1 r + a_2 r^2$	$R^2 \approx 0.95$	EP-04 ENSDF Pb-206
Master Buoyancy	$U_{b, Master} = U_{b,i} + e^{-(\pi - t)} U_m / \rho_{vac, [UA]}$	π -alignment	Full TRZ cycles

References

Grok thread 2fe4fa3e (Sept 22, 2025). DeepSearch extraction: 7 UQFF equation systems with variable tables.

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k_i	$k_1=1.5, \backslash; k_2=1.2, \backslash; k_3=1.8, \backslash; k_4=1.0$	■	Coupling constants
$U_{\{g1\}}$	$k_1 \backslashmu_s (M_s/r) e^{\{-\alpha t\}} \backslashcos(\pi t_n)(1+\backslashb\eta_{\{def\}})$	N/m■	Dipole gravity
$U_{\{g2\}}$	$k_2(\backslashlambda_{\{vac, [UA]\}}+\backslashlambda_{\{vac, [SCm]\}})(M_s/r^2)\backslash, S(r-R_b)(1+\backslashdelta_{\{sw\}}v_{\{sw\}})H_{\{SCm\}}E_{\{react\}}$	N/m■	Bubble gravity
$U_{\{g3\}}$	$k_3 \backslashsum_j B_j \backslashcos(\omega_s t)\backslash, P_{\{core\}}\backslash, E_{\{react\}}$	N/m■	Disk strings gravity
$U_{\{g4\}}$	$k_4 \backslashlambda_{\{vac, [SCm]\}}(M_{\{bh\}}/d_g)e^{\{-\alpha t\}}\backslashcos(\pi t_n)(1+f_{\{feedback\}})$	N/m■	BH-star interaction
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$\backslasheta$	1×10^{-22}	■	Aether coupling
$T_s^{\{\mu\nu\}}$	$\backslashapprox 1.123\times10^7$ J/m■	J/m■	Stress-energy tensor

Symbol	Value / Equation	Units	Description
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U_i	$\lambda_i \rho_{vac, [SCm]} \rho_{vac, [UA]} \omega_s(t) \cos(\pi t_n) (1 + f_{TRZ})$	J/m■	Universal inertia
D_E	$\propto E^{0.5}$ (Kolmogorov)	■	CRP diffusion coefficient
$n(p)$	$p^{-2.2} e^{-p/p_{max}}$	m?■ (GeV/c)?■	CRP momentum distribution

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γ	0.00005 day?■	day?■	Decay rate, $\tau = 1/\gamma \approx 54.8$ yr
κ	0.0005 day?■	day?■	Reactivity decay, $\tau_{\kappa} \approx 5.48$ yr
$(1 - e^{-\gamma t} \cos(\pi t_n))$	$\approx \gamma t \cos(\pi t_n)$ for small t	■	Resonant buildup in U_m
f_{TRZ}	0.1	■	TRZ factor scaling U_i for negentropy
ω	π rad/s (rad/day)	rad/day	Cycle constant

Application: EP-09 (3C 273 jet ratio $R = |\cos(\pi t_{n1})/\cos(\pi t_{n2})| > 100$ for $\Delta t \approx 0.5$ days).

4. System 3: UQFF Buoyancy (Ub_i Opposition System)

Long Form:

$$U_{b,i} = -\beta_i U_{g,i} \cdot \frac{\omega_g M_{bh}}{d_g} \cdot (1 + \delta_{sw} \lambda_{vac,sw}) \cdot [UA] \cdot \cos(\pi t_n)$$

Variable Equations:

Symbol	Value / Equation	Units	Description
$U_{\{b,i\}}$	Full expression above	J/m^4	Buoyancy density (opposes $U_{\{g,i\}}$)
β_i	0.61	$\frac{m}{m^4}$	Buoyancy coupling (uniform $i=1-4$)
$U_{\{g,i\}}$	See System 1	J/m^4	Gravity component
ω_g	7.3×10^{-16} rad/s	rad/s	Galactic rotation rate
$M_{\{bh\}}$	8.15×10^{36} kg	kg	SMBH mass (Sgr A*)
d_g	2.55×10^{20} m	m	Galactic distance
$\delta_{\{sw\}}$	0.01	$\frac{m}{m^4}$	Solar wind modulation factor
$\lambda_{\{vac,sw\}}$	$\frac{\rho_{\{sw\}}}{c^2} \approx 7.2 \times 10^{-4} J/m^4$	J/m^4	Wind vacuum density
[UA]	10^{-11} C	C	Trapped Aether charge
$\cos(\pi t_n)$	Oscillator	$\frac{m}{m^4}$	Time-reversal modulation

Physical interpretation: $U_{\{b,i\}}$ feeds outflows by opposing gravity. For BNS mergers (EP-11, GW170817): dynamical ejecta fraction $\approx 1 - \beta_i \approx 0.39 \approx 40\%$. For jets: $\cos(\pi t_n)$ reversals create $R > 100:1$ asymmetry (EP-09).

5. System 4: UQFF Superconductive ([SCm] Reactivity)

Long Form (dual form):

$$E_{\{react\}} = 10^{46} e^{-0.0005 t} \equiv \frac{\rho_{\{vac,[SCm]\}}}{v_{\{SCm\}}^2} \rho_{\{vac,A\}} e^{-\kappa t}$$

Variable Equations:

Symbol	Value / Equation	Units	Description
$E_{\{react\}}$	$10^{46} e^{-\kappa t}$	W/m^4	[SCm] superconducting reactivity
$\rho_{\{vac,[SCm]\}}$	$7.09 \times 10^{-37} J/m^4$	J/m^4	SCm vacuum energy density
$v_{\{SCm\}}$	1×10^8 m/s (= c/3)	m/s	SCm propagation velocity
$\rho_{\{vac,A\}}$	$1 \times 10^{-23} J/m^4$	J/m^4	Aether vacuum density
κ	0.0005 day $^{-1}$	day $^{-1}$	Reactivity decay constant
10^{46}	$= \frac{\rho_{\{vac,[SCm]\}}}{v_{\{SCm\}}^2} \rho_{\{vac,A\}}$	W/m^4	Quasar-scale base efficiency
[SCm]	10^{15}	m $^{-1}$	Superconductive material density
[UA]	10^{-11}	C	Universal Aether charge

Calibration: EP-05 (Fermi LAT 4LAC blazars): $\bar{\kappa} = 0.000497 \pm 5\% \text{ day}^{-1}$ across 3,743 sources. EP-07 (Parker Solar Probe): $\Delta_{\text{sw}} = 0.01 = [\text{SSq}]/57$.

6. System 5: UQFF Triadic (Triadic Master Equations)

Long Form (e.g., Westerlund 2 plasma turbulence):

$$F_{\text{U, Triadic}} = F_{\text{U}} + \left(U_{\text{g3}} \cdot U_{\text{b,i}} \cdot U_{\text{m}} \right)^{1/3} \cdot \exp \left(- [\text{SSq}] \cdot \frac{n}{26} \right)$$

Variable Equations:

Symbol	Value / Equation	Units	Description
$F_{\text{U, Triadic}}$	F_{U} + geometric mean term	J/m ³	Triadic unified field
$(U_{\text{g3}} \cdot U_{\text{b,i}} \cdot U_{\text{m}})^{1/3}$	Geometric mean of disk gravity, buoyancy, magnetism	J/m ³	Triadic coupling
[SSq]	$\log_{10}(\rho_{\text{vac}} / \lambda_{\text{vac}}) \approx 38$ (for 10 ⁻³⁸ ratios)		Self-similar quotient
n	1-26 (level index)		UQFF ladder level
n = 13	Plasma systems (Westerlund 2, Pillars)		Triadic plasma level
$\exp(-[\text{SSq}] \cdot n/26)$	$\exp(-38 \cdot n/26)$		Level-scaled suppression
$Q_{\text{wave}, 47}$	Mean: 3.97×10 ⁴ J/m ³ , std: 5.11×10 ⁴ J/m ³	J/m ³	47-system wave energy density
Jarque-Bera	8.78 (p=0.012)		Non-normality statistic for Q _{wave}
Leptokurtosis	0.037		Excess kurtosis of Q _{wave} distribution

Systems using Triadic: Westerlund 2 (turbulence), Pillars of Creation (buoyancy), TOI 1227 b (mass loss $\approx 10^{12}$ g/s), PLCK G287.0+32.9 (double relics), ASKAP J1832-0911, PSZ2 G181.06+48.47, AT2024tvd, G359.13142-0.20005.

7. System 6: UQFF Quadratic (Quadratic Field Approximations)

Long Form (low-order approximation of potential):

$$V(r) \approx a_0 + a_1 r + a_2 r^2$$

$$T_s^{\mu\nu} = 1.27 \times 10^3 + 1.11 \times 10^7 \text{ J/m}^3 \text{ (quadratic sum)}$$

Variable Equations:

Symbol	Value / Equation	Units	Description
$V(r)$	$\sum_n a_n r^n$ approx quadratic for low deg	J or m/s	Field potential approximation
a_0	Constant term		Polynomial fit coefficient 0
a_1	Linear coefficient		Polynomial fit coefficient 1
a_2	Quadratic coefficient (approx 10^{-12} for n=8)		Polynomial fit coefficient 2
R^2	approx 0.95 (for ENSDF n=8 bindings)		Quadratic fit quality
$T_s^{\{\mu\nu\}}$	$1.27 \times 10^3 + 1.11 \times 10^7$	J/m	Stress tensor quadratic sum
$a_2^{\{\text{Pb-206}\}}$	approx 10^{-12}		Fit for EP-04 Pb-206 n=8 levels

Application: Nuclear binding energy polynomial fits (EP-04 ENSDF Pb-206, EP-02 PDG 2025) use degree-8 polynomials that reduce to effective quadratic forms for low-n regimes.

8. System 7: UQFF Master Buoyancy (Extended $U_{b,i}$ with Mayan Alignment)

Long Form:

$$U_{\{b, \text{Master}\}} = -\beta_i U_{\{g, i\}} \frac{\omega_g}{M_{\{bh\}}\{d_g\}(1+\delta_{\{sw\}}\lambda_{\{vac, sw\}})[UA]\cos(\pi t_n) + \exp(-(p - t)) \cdot \frac{U_m}{\rho_{\{vac, [UA]\}}}$$

Physical basis: Extends standard $U_{\{b, i\}}$ with a time-scaled magnetic feed term. The $\exp(-(p-t))$ factor aligns with Mayan calendar time constants (Baktun cycle) in the UQFF cosmological scaling.

Variable Equations:

Symbol	Value / Equation	Units	Description
$U_{\{b, \text{Master}\}}$	Full expression above	J/m	Master buoyancy extended form
Standard $U_{\{b, i\}}$	System 3 above	J/m	Base buoyancy term
$\exp(-(\pi - t))$	For $t < \pi$ days, decays toward Euler		Mayan/Master time alignment factor
U_m	$\sum_j [\mu_j / r_j (1 - e^{-\gamma t \cos(\pi t_n)}) \hat{\phi}_j]$ $P_{\{SCm\}} E_{\{react\}}$	J/m	Universal magnetism (lossless strings)
$\rho_{\{vac, [UA]\}}$	7.09×10^{-36} J/m	J/m	UA vacuum energy density
$P_{\{SCm\}}$	approx 1 (Sun)		SCm core penetration factor
$E_{\{react\}}$	$10^{46} e^{-\kappa t}$	W/m	Reactor efficiency

****Distinction from PAPER_064:**** PAPER_064 documents the four simplified operational modes. This Master Buoyancy form adds the resonant magnetic feed term $e^{-(p-t)} U_m / \rho_{\{vac, [UA]\}}$, which

drives enhanced negentropy in long-period TRZ cycles spanning Mayan-scale time constants (Baktun = 394 yr ■ $1/\kappa^{0.33}$).

9. Shared Constants Across All Seven Systems

Constant	Value	Physical Meaning
κ	0.0005 day?■	E _{react} decay rate (EP-05 Fermi LAT)
[SSq]	0.57	Self-similar quotient (EP-04 Pb-206 S _n)
β_i	0.61	Buoyancy coupling (EP-10 IceCube, EP-11 GW170817)
F _{rel}	4.31×10 ³³ N	Relativistic unified force scale
k _{eta}	10 ⁻¹¹³	LENR exponential damping factor
ω	π rad/day	UQFF cycle constant
α	0.001 day?■	Dipole/BH time decay in U _{g1} , U _{g4}
δ_{sw}	0.01 = [SSq]/57	Solar wind modulation (EP-07 PSP)
v _{sw}	5×10 ⁵ m/s	Solar wind velocity (EP-07)
H _{SCm}	approx 1	Heliosphere SCm factor
f _{feedback}	0.1	AGN/BH feedback factor
f _{TRZ}	0.1	TRZ negentropy factor
λ_i	1.0	Inertial coupling for U _i
p _{max}	10 ¹⁶ eV	CRP momentum cutoff (EP-10 IceCube)
γ_{CRP}	0.00005 day?■	CRP decay rate (τ approx 55 yr)

10. Cross-Reference to Implemented Code

System	Source Code	Class/Function	PAPER_064 equiv.
Compressed	source2.cpp L1960, add_uqff_to_8_models.py L24	UQFF_Compressed	Mode 1 ?
Resonant	source2.cpp L1961, add_uqff_methods.py	UQFF_Resonant	Mode 2 ?
Buoyancy	add_uqff_to_8_models.py L68	F_U_Bi_i / UQFFMasterBuoyant	Mode 3 ?
Superconductive	add_uqff_to_8_models.py L48	UQFF_Superconductive	Mode 4 ?
Triadic	add_uqff_to_8_models.py L76, add_uqff_methods.py L226	UQFF_Triadic	■ (new)
Quadratic	add_uqff_to_8_models.py L90, add_uqff_methods.py L291	UQFF_Quadratic	■ (new)
Master Buoyancy	add_uqff_to_8_models.py L68	UQFFMasterBuoyant	■ (new)

11. Summary Table

System	Core Equation	Key Parameter	Primary Verification
Compressed	$F_U = \sum_i k_i U_{gi} - \beta_i U_{gi} \omega_g M_{bh}/d_g E_{react}$	$\kappa = 0.0005 \text{ day}^{-1}$	EP-05 Fermi LAT
Resonant	$\cos(\pi t_n), (1 - e^{-\gamma t} \cos(\pi t_n))$	Period = 2 days	EP-09 3C 273
Buoyancy	$U_{b,i} = -\beta_i U_{g,i} \omega_g M_{bh}/d_g (1 + \delta \lambda_{vac,sw}) [UA] \cos(\pi t_n)$	$\beta_i = 0.61$	EP-11 GW170817
Superconductive	$E_{react} = 10^{46} e^{-\kappa t}$	$\kappa = 0.0005 \text{ day}^{-1}, v_{SC} = c/3$	EP-08 JCAP
Triadic	$F_{U,tri} = F_U + (U_{g3} U_{b,i} U_m)^{1/3} e^{-[SS]n/26}$	$[SS] = 0.57, n = 13$	Qwave47 mean
Quadratic	$V(r) \approx a_0 + a_1 r + a_2 r^2$	$R^2 \approx 0.95$	EP-04 ENSDF Pb-206
Master Buoyancy	$U_{b,Master} = U_{b,i} + e^{-(\pi - t)} U_m / \rho_{vac, [UA]}$	π -alignment	Full TRZ cycles

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